

CARLOS SEMEHO EDORH

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EDUCATION

Northeastern University

Vancouver, BC

M.S. in Computer Science

Graduated, December 2025

- **Concentrations:** Software Engineering
- **Current GPA:** 3.79/4.00
- **Related Coursework:** Web Development, Mobile Application Development, Algorithms, Building Game Engines, Data Mining Techniques, Computer/Human Interaction, Mobile Cyber Security.

University of Education, Winneba

Ghana

Bachelor of Science in Information and Communication Technology

Graduated: November 2014

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, PHP, SQL, C++, C#, HTML5, CSS3, Python, Kotlin

Frontend: React.js, Next.js, React Native, Redux, Tailwind CSS, Bootstrap

Backend: Node.js, Express.js, RESTful APIs, Drizzle & Prisma ORM

Databases: MySQL, MongoDB, Firebase, SQLite, Microsoft SQL Server

Frameworks & CMS: WordPress, CodeIgniter, .NET Framework

Tools & Platforms: Git, GitHub, cPanel, Expo, EAS Build, Stripe, Twilio, SMS Gateways, Visual Studio, VS Code

Methodologies: SOLID Principles, Agile/Scrum, MVC Architecture, ECS Pattern, Test-Driven Development

EXPERIENCE

Khoury College of Computer Science – Northeastern University

Vancouver, BC

Teaching Assistant for CS 5610 (Web Development)

Jan 2025 – Apr 2025

AECI Explosives (formerly AEL Intelligent Blasting)

Western Region, Ghana

Software Developer

Sep 2017 – Aug 2024

- Developed 6+ internal systems serving 200+ employees: vehicle management, safety alerts, production tracking, pit movements, archive management, and inventory systems.
- Developed custom middleware integrations connecting SAP ERP with third-party systems including TurnStar(Morpho) Biometric Access Control and ABLOY High Security Padlock cylinders, reducing manual processes by 60%.
- Collaborated with cross-functional teams and stakeholders to gather requirements, architect solutions, and deliver projects aligned with business objectives by digitizing some existing manual process listed above.
- On recommendation from AECI, provided specialized services to partner mining firms, including **AngloGold Ashanti (Obuasi)** for ABLOY High Security Padlock systems implementation, **ADAMUS Resources (Nzema)** for Archive Management System design, and **Gateway Logistics (Takoradi)** as a translator for training French-speaking drivers on safe Ammonium Nitrate handling during transport, **Magnum Security Force (Accra)** as a remote e-Document System and Email system Manager.

Freelance Software Developer

Ghana

Contract & Independent Projects

Aug 2015 – Aug 2024

- Developed multiple client projects including **E-Recruitment System**, **Employee Insurance Management**, **School Management Systems(3 versions)**, **Barber Shop Management**, **Savings & Loans**, e-commerce platforms, and logistics solutions

- Delivered annual IT training programs (3 months, August-November) to 500+ Ghana Prison Service personnel across 3 regions (Volta, Eastern, Greater Accra), covering Microsoft Office Suite, database management, and web technologies from 2020-2022

University of Education, Winneba

Teaching Assistant (National Service)

Ghana

Aug 2014 – June 2015

Assisted in teaching undergraduate ICT courses, conducted programming lab sessions, and evaluated student assignments.

GRDUATE PROJECTS

ShieldShare - VPN-backed Android Hotspot with Traffic Accounting | Android, VpnService API, Proxy Server
CS 7980: Research Capstone in Cybersecurity (Fall 2025)

Developing an Android app enabling secure hotspot sharing through VPN tunnels using proxy-based architecture; includes per-user data metering and security testing for traffic leakage.

TimeOasis - Leave Management SaaS Platform | React, Node.js, Express, MySQL, Prisma ORM
CS 5610: Web Development (Fall 2024)

Built full-stack SaaS platform with role-based access, JWT authentication, and Stripe billing; deployed secure RESTful APIs and automated workflows for multi-tenant companies.

Local Buzz - Hyperlocal Community App | React Native, Firebase, Google Cloud AI, Expo - CS 5520: Mobile Application Development (Spring 2025)

Created cross-platform mobile app featuring real-time community updates, geospatial filtering, and AI-driven content categorization; integrated Firebase for synchronization and notifications.

Pacman ECS Game Engine | TypeScript, Canvas API, Box2D Physics, ECS Architecture - CS 5850: Building Game Engines (Summer 2025)

Developed a 2D game engine with Entity Component System architecture and real-time collision detection; implemented modular animation, physics, and audio systems.

ACTIVITIES

Campus Volunteer

Student

Vancouver, BC

Jan 2025 –Current

Supported career fairs, student orientations, and campus events

CERTIFICATION & AWARDS

MINT Robotics Training Workshop (Company Certified) – Ghana, 2011

WHMIS 2015 (Workplace Hazardous Materials Information System) - Canada Safety Training, 2025

Accident and Incident Investigation Training Certification - Canada Safety Training, 2025

Workplace Health and Safety Awareness Certificate - Canada Safety Training, 2025

LANGUAGE

Languages: English (Fluent), French (Fluent)